

MASTER OF SCIENCE ARTIFICIAL INTELLIGENCE & MACHINE LEARNING (MSCAIML)

MSCAIML/ASSIGN/SEMESTER-I

ASSIGNMENTS

(July – 2026)

**MCS-061, MCS-081, MCS-208, MCS-082, MCSL-209,
MCSL-083**



**SCHOOL OF COMPUTER AND INFORMATION SCIENCES
INDIRA GANDHI NATIONAL OPEN UNIVERSITY
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Important Notes

1. Submit your assignments to the Coordinator of your Study Centre on or before the due date.
2. Assignment submission before due dates is compulsory to become eligible for appearing in corresponding Term End Examinations. For further details, please refer to MSCAIML Programme Guide.
3. To become eligible for appearing in the Term End Practical Examination for the labcourses, it is essential to fulfill the minimum attendance requirements as well as submission of assignments (on or before the due date). For further details, please refer to the MSCAIML Programme Guide.
4. The viva voce is compulsory for the assignments. For any course, if a student submitted the assignment and not attended the viva-voce, then the assignment is treated as not successfully completed and would be marked as ZERO.

Course Code : **MCS-082**
Course Title : **Programming Using Python**
Assignment Number : **MSCAIML(I)/082/Assign/2026**
Maximum Marks : **100**
Weightage : **30%**
Last Dates for Submission : **31st October, 2026 (for July 2026 Session)**

Note: There are eight questions in this assignment, carrying a total of 80 marks. The remaining 20 marks are for the viva voce. Answer all the questions. Give the running code and its output as part of your answer wherever required.

- Q1: (a)** Compare and contrast lists, tuples, and sets in Python in detail, and provide an example Python script to illustrate your answer. **(5 Marks)**
- (b)** Explain how indexing and slicing mechanics function on multi-dimensional list elements. Write a program containing a nested list (a matrix) and use slice syntax variations to: **(5 Marks)**
1. Extract a sub-matrix grid block.
 2. Reverse the order of elements in individual inner lists using in-place reassignments.
- Q2: (a)** Discuss the procedural execution behaviour of loop variations in Python. Differentiate between while loops and for loops, and write a program that demonstrates how the break and continue keywords disrupt standard sequential control flow paths. **(5 Marks)**
- (b)** Describe the operational hierarchy of Pandas Series and DataFrame objects. **(5 Marks)**
- Q3: (a)** Analyze how Python processes variable names globally and locally, using example code to support your explanation. **(5 Marks)**
- (b)** Contrast regular named function architectures with lambda (anonymous) functions. Write a program demonstrating a higher-order function that accepts a lambda expression as an input parameter to filter and map a dictionary dataset based on custom key conditions. **(5 Marks)**
- Q4: (a)** Explain the concepts of an instance method, a class method (@classmethod), and a static method (@staticmethod) in Python's OOP model, providing examples for each. **(6 Marks)**
- (b)** Explain the structural execution mechanics of *decorators* and *generators* inside modern Python runtime engines. **(4 Marks)**
- Q5: (a)** Define polymorphism in Python and explain how operator overloading enables custom behaviours for standard operand symbols. **(5 Marks)**
- (b)** Describe the operational flow of Python's complete exception-handling block architecture (try, except, else, and finally) with the help of a program. **(5 Marks)**
- Q6: (a)** What is multithreading, and how does Python support it? Provide a foundational script utilising the threading module that synchronizes two overlapping threads competing for a single shared resource using thread-primitive lock systems. **(5 Marks)**

- (b) Write Python code utilising Tkinter that constructs a user login window consisting of entry input containers, tracking label grids, and an active submission command trigger button. **(5 Marks)**

Q7: (a) Explain the structural differences between the Flask and FastAPI web framework ecosystems for developing enterprise services. Also, explain the concepts of synchronous routing blocks versus asynchronous event-handling loops (async/await), and describe how Pydantic validation models help in automated data verification. **(5 Marks)**

- (b) Explain the operational lifecycle required to safely open, execute, commit, and terminate transactions on relational database engines using Python. Also, explain how developers protect database query interfaces against SQL injection vulnerabilities. **(5 Marks)**

Q8: (a) Explain why NumPy Array (ndarray) calculations dramatically outperform standard nested native Python lists for heavy data calculations. Furthermore, explain what vectorization and broadcasting rules imply when processing multi-dimensional array indices. **(5 Marks)**

- (b) Explain the uses of Matplotlib visualization charts, including the explicit structural hierarchy among a Figure, an Axes object class, and an individual Axis. Make necessary assumptions. **(5 Marks)**